Answer Key & Teacher Notes

Session 1 — Statistical Questions · Session 2 — Dot Plots and Shape

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Statistical Questions

TALK PROMPTS

- How can you tell, before collecting any answers, whether a question will produce variability?
- Five shoppers gave five different amounts. How does that list show variability?

PARTNER TASK

Reading Minutes. One partner reads a question aloud; the other predicts three answers classmates might give. If all three must match, the pair rewrites the question so the answers vary. Switch roles.

WATCH FOR

- Thinking any question with a number answer is statistical ("How many minutes are in an hour?").
- Calling a question about one person or one object statistical ("How tall is the flagpole?").
- Skipping 0 when finding the least answer in a data set.

SESSION 2 · TEACHER NOTES

Dot Plots and Shape

TALK PROMPTS

- How does a dot plot show at a glance which answer came up most often?
- What does one single dot on your plot stand for?
- How would the shape change if one more student answered the greatest value?

PARTNER TASK

Library Books. Partners plot the ten book counts together, one placing dots and one checking the count (10 dots). Then each predicts how the shape changes if one more student finishes 5 books, and they test it.

WATCH FOR

- Drawing one dot per different value instead of one dot per data value.
- Naming skew by where the dots pile up instead of by the tail.
- Reading the height of the tallest stack as the most common value (4 dots at 2 means the value is 2).

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.SP.A.1	1	1	1	2	1	2	2	3	2	1	1
S1 · Form B	6.SP.A.1	1	1	1	2	1	2	2	3	2	1	1
S2 · Form A	6.SP.B.4	1	1	1	1	2	2	2	2	2	1	1
S2 · Form B	6.SP.B.4	1	1	1	1	2	2	2	2	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

statistical question: A question whose answers are expected to **vary**.

variability: How **different** the answers in a data set are.

data: Facts or **numbers** you collect to answer a question.

- A statistical question expects answers that **vary**, so you collect data to answer it.
- A question with one **fixed** answer is not statistical, even if the answer is a **number**.

Example: Who answers? **Each** student. Will the answers differ? Yes: 5, 12, 25 minutes, so the answers **vary**. Answer: The question is **statistical**.

YOU TRY

1. a. N b. S

2. Model: "How tall is each student in our class?"

EXIT TICKET

1. a. N b. S

2. Model: "How many people live in each student's home?"

3. **C** (How many minutes did each student read?)

PRACTICE FORM A

1. a. S b. N c. N d. S

2. a. S (the answers vary) b. 6 c. 0 d. 40

3. Question 1. Each student gives a different answer; question 2 has one fixed count.

4. No, not statistical. Everyone counting the same tables gets one number, so there is no variability.

5. a. S (amounts vary) b. 7 dollars c. 22 dollars

6. Model: "How many years has each student attended our school?" Each student answers about themselves, so the answers vary.

7. a. N b. S c. S; 2 are statistical

8. a. Model: "How many hours did each sixth grader sleep last night?" b. The number of hours varies from student to student. c. It asks about one person, so it has one answer and no variability.

9. Having a number answer is not enough. There are always 60 minutes in one hour, so the answer never varies; it is not statistical. Model: "How many minutes does each student spend on homework?"

10. **D** (How old is each teacher at our school?)

11. **C** (variability)

Self-check word **SURVEY**

PRACTICE FORM B

1. a. S b. N c. N d. S

2. a. S (the answers vary) b. 8 c. 5 d. 50

3. Question 1. Each student gives a different answer; question 2 has one fixed count.

4. No, not statistical. Everyone counting the same windows gets one number, so there is no variability.

5. a. S (amounts vary) b. 9 dollars c. 35 dollars

6. Model: "How many floors does each student's home have?" Each student answers about their own home, so the answers vary.

7. a. S b. N c. S d. S; 3 are statistical

8. a. Model: "How many hours did each sixth grader sleep last night?" b. The number of hours varies from student to student. c. It asks about one person, so it has one answer and no variability.

9. Having a number answer is not enough. There are always 12 inches in one foot, so the answer never varies; it is not statistical. Model: "How many inches long is each student's foot?"

10. **C** (What is my locker number?)

11. **A** (are expected to vary)

Self-check word **ANSWER**

GUIDED NOTES

dot plot: A number line with one **dot** for each data value.

peak: The **tallest** stack; it shows the most common value.

symmetric: The left and right sides of the plot **match**.

skewed: The data has a long **tail** on one side.

- Repeated values **stack** up, so the total number of dots equals the number of **data values**.
- A plot is named for its **tail**: a tail to the left means skewed left.

Example: Draw a line from 3 to 8 and label each whole number. Stack one dot per friend: **3** dots above 5. The peak is at **5**; there are **6** dots in all. Answer: Most common value: **5** grapes

YOU TRY

1. 8; symmetric

2. a. 10 b. 4 dots

EXIT TICKET

1. a. 3 b. 6

2. Skewed left (peak at 3, tail toward 1)

3. **B** (10)

PRACTICE FORM A

1. Dots: 1 (1), 2 (4), 3 (2), 4 (1), 5 (1). a. 2 b. skewed right (the peak is at 2 and the tail stretches right to 5)

2. Dots: 6 (3), 7 (1), 8 (1), 9 (1), 10 (1). Most common: 6

3. a. 5 b. 7

4. 10

5. Yes. The stacks are 2, 3, 2: the sides match around the peak at 8.

6. Most common: 12 (stacks 3, 2, 2); the greatest value stays 18.

7. Most common: 11; skewed left (the peak is at 11 and the tail stretches left to 9).

8. Dots: 1 (1), 2 (3), 3 (4), 4 (1), 5 (1). Most common: 3

9. Marcus drew 3 dots for 4 values. The value 3 appears twice, so it needs a stack of 2 dots. The plot should show 2 dots above 3 and 1 dot above each other value.

10. **B** (how many times that value appears)

11. **A** (over 5)

Self-check word **STACKED**

PRACTICE FORM B

1. Dots: 1 (1), 2 (1), 3 (4), 4 (2), 5 (1). a. 3 b. nearly symmetric (stacks 1, 1, 4, 2, 1 balance around 3)

2. Dots: 5 (3), 6 (1), 7 (1), 8 (1), 9 (1). Most common: 5

3. a. 6 b. 9

4. 12

5. Yes. The stacks are 2, 3, 2: the sides match around the peak at 5.

6. Most common: 20 (stacks 3, 2, 2); the greatest value stays 28.

7. Most common: 16; skewed left (the peak is at 16 and the tail stretches left to 14).

8. Dots: 2 (1), 3 (3), 4 (4), 5 (1), 6 (1). Most common: 4

9. Eli drew 3 dots for 4 values. The value 5 appears twice, so it needs a stack of 2 dots. The plot should show 2 dots above 5 and 1 dot above each other value.

10. **C** (most common value)

11. **A** (1, 2, 2, 3)

Self-check word **PLOTTED**

SHEET 1 · STATISTICAL QUESTIONS

1. a. N b. S c. N d. S
2. Statistical. The answers (4, 5, 6, 7) are different, so the data vary.
3. Statistical. Each player lives a different distance away, so the answers vary.
4. It has one answer (one tree, one height). Model rewrite: "How tall is each tree in the park next to our school?"
5. Model: "How many hours did each person sleep last night?" Possible answers: 6, 8, 9 hours (the answers vary).

SHEET 2 · DOT PLOTS AND SHAPE

1. Dots: 1 (one), 2 (three), 3 (four), 4 (three), 5 (one). a. 3 b. Symmetric (both sides match).
2. Dots: 2 (one), 3 (one), 4 (two), 5 (four), 6 (one). a. 4 b. 5 tickets c. Skewed left (tail toward 2)
3. a. 11 games b. 1 goal c. Between 4 and 7 (no games with 5 or 6 goals)
4. Skew is named for the tail, not where the dots pile up. The tail stretches right toward 7 (an outlier), so the plot is skewed right.
5. Skewed left. Most scores are high (peak at 9); the tail stretches left toward 5.
6. **A** (Symmetric)
7. **D** (15)

October 2026 conceptual additions — teacher key

Session 1 · Statistical Questions

A is statistical: the variable is daily running time in minutes, measured over six days, and times may vary. B requests one recorded value. A statistical investigation need not involve many different people.

Extension: Example: What are the lengths, in centimeters, of the pencils in this box? Population: pencils in the box; variable: length; unit: centimeters; lengths may vary.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Dot Plots and Shape

One dot at 2, three at 3, one at 5, one at 6; six dots total. Peak: 3 books. Gap within the observed range: 4 books. Each observation contributes exactly one dot.

Extension: A single added observation cannot tie another value with the frequency 3 at the existing peak: frequencies at 2, 5, or 6 rise only to 2. Students should explain this impossibility rather than invent a second peak.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.